

Northern Lights — Open-Access CO₂ Transport and Storage Service

Building the World's First CCS Infrastructure as a Service

QUICK FACTS

- Sector:** CO₂ Transport and Storage Infrastructure
- Location:** Oygarden, Norway (onshore terminal) + North Sea (offshore storage)
- Technology:** Ship-based liquid CO₂ transport; subsea pipeline; Johansen saline aquifer injection

1. Background: The Infrastructure Problem in CCS

One of the most significant barriers to widespread CCS deployment is not technology — it is infrastructure. A cement plant in Rotterdam wanting to capture its CO₂ faces a daunting question: where does it go? Building a dedicated offshore pipeline and developing a geological storage site for one plant is economically prohibitive. The capital cost is enormous; the risk is undiversified; and the geology must be proven from scratch.

Northern Lights solves this problem by separating the transport and storage function from the capture function — creating a utility-style service that any emitter can access, rather than requiring each emitter to develop its own bespoke solution.

Northern Lights is a joint venture (JV) between Equinor (51%), Shell (25%), and TotalEnergies (24%), established under Norway's Longship CCS programme with substantial Norwegian government financial support.

JV partners	Equinor (51%), Shell (25%), TotalEnergies (24%)
Onshore terminal	Oygarden, Hordaland county, western Norway
Terminal function	CO ₂ receiving, temporary storage, re-pressurisation
Offshore pipeline	~100 km subsea pipeline from Oygarden to injection site
Storage formation	Johansen sandstone, 2,600 m below North Sea seabed
Phase 1 capacity	1.5 million tonnes CO ₂ /year (2024 onwards)
Phase 2 capacity (planned)	5 million tonnes CO ₂ /year
Initial customers	Norcem Brevik (cement) + Hafslund Oslo Celsio (waste-to-energy)
Business model	Open-access CO ₂ transport and storage service; tariff-based

Norwegian govt support	~60% of capital cost (Phase 1) through Longship programme
Total Phase 1 investment	~NOK 7 billion (~USD 650 million)

2. The Business Model: Infrastructure as a Service

Northern Lights's business model is deliberately designed as a utility service, analogous to how electricity grid operators or gas pipeline companies function. This design choice has profound implications for CCS deployment:

Business Model Feature	Description	Why It Matters for CCS Deployment
Open access	Any emitter can subscribe; no exclusivity	Removes the need for each emitter to develop own infrastructure
Long-term contracts	10–25 year CO ₂ transport and storage agreements	Provides revenue certainty; enables project finance
Tariff-based revenue	Customers pay per tonne CO ₂ transported and stored	Simple, transparent pricing; independent of customer's own economics
Phase-based expansion	Phase 1: 1.5 Mt/year; Phase 2: 5 Mt/year	Capacity added as customers are secured; reduces stranded asset risk
Government support	Norwegian government funded ~60% of Phase 1 capital	De-risks first-of-a-kind infrastructure; creates platform for private investment in Phase 2
Shared storage	Multiple customers inject into same formation	Reduces per-customer storage characterisation cost dramatically

3. The Johansen Storage Formation

The Johansen sandstone formation is one of the most thoroughly characterised offshore CO₂ storage sites in the world. It was first evaluated in the 1990s as a potential storage site for the Sleipner project, though ultimately not selected for that project. Decades of subsequent study have confirmed its suitability:

Property	Johansen Formation
Depth below seabed	~2,600 m
Formation thickness	~100 m (net sand)
Porosity	~20–25%
Permeability	~100–500 mD

Property	Johansen Formation
Caprock	Dunlin shale; 50–100 m thick; proven integrity
Estimated storage capacity	80–600 million tonnes CO ₂ (Johansen cell)
Risk of leakage	Very low; multiple trapping mechanisms; deep burial

4. Phase 2 Expansion and European CCS Market Development

Phase 2 of Northern Lights aims to expand capacity to 5 million tonnes per year, serving customers from across Europe. The potential customer base is substantial:

- **Waste-to-energy:** Europe has ~500 waste incineration plants, many in coastal cities. Their flue gas contains significant biogenic CO₂ — capturing it creates net-negative credits. Several are in active discussions with Northern Lights.
- **Cement:** Multiple European cement plants beyond Norcem are evaluating CCS. Plants in Belgium (HeidelbergMaterials), Germany, and the Netherlands are priority targets.
- **Hydrogen production:** Blue hydrogen production facilities using CCS, primarily in Germany and the Netherlands, could route CO₂ to Northern Lights via pipeline to the coast.
- **Direct Air Capture:** Early-stage DAC projects may use Northern Lights for permanent storage of their captured CO₂.

5. Lessons and Relevance for India

- **Shared infrastructure is the key to cost reduction:** Northern Lights demonstrates that separating the storage function from the capture function — and sharing it across multiple emitters — reduces the per-tonne cost of geological storage by 50–70% compared to bespoke single-emitter solutions.
- **Government investment unlocks private capital:** Norway's 60% capital contribution to Phase 1 was essential. No private sector consortium would have taken the full financial risk for a first-of-a-kind infrastructure project. Government de-risking, followed by private investment in Phase 2, is the right sequencing model.
- **India's coastal industrial clusters:** Odisha, Andhra Pradesh, Gujarat — India's industrial coasts have significant CO₂ emitter concentrations. A shared CO₂ infrastructure model, with Indian government initial investment and private sector Phase 2, could serve multiple emitters at a fraction of the cost of individual solutions.
- **Storage characterisation first:** Northern Lights works because the geology was known. India must invest in offshore geological storage characterisation now — a 5–10 year process — to enable CCS infrastructure development in the 2030s.

Key Takeaway: Northern Lights is not just a CCS project. It is a proof-of-concept for a new model of climate infrastructure — shared, open-access, utility-scale. This model can and should be replicated in India.